

Qualification Pack



Junior Engineer (JE)-Power Distribution

QP Code: PSS/Q3004

Version: 1.0

NSQF Level: 5

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PSS/Q3004: Junior Engineer (JE)-Power Distribution

Brief Job Description

Supervise and carry out installation, operation & maintenance, testing & inspection of 33/11 Substation to last mile consumer supply including distribution transformer.

Personal Attributes

Work is performed indoors as well as outdoors in all weather conditions. Work requires the ability to perform engineering and coordination activities in the work place. Work also involve bending, walking, and standing for significant periods of time. Candidate will be exposed to different types of power supply areas and irregular terrain. Periodic night-time work is also required.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1336: Working effectively with others](#)
2. [PSS/N2001: Use basic health and safety practices for power related work](#)
3. [PSS/N3007: Carry out Installation in power distribution systems](#)
4. [PSS/N3008: Carry out operation and maintenance of power distribution system](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
Country	India
NSQF Level	5
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2004/NIL



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Minimum Educational Qualification & Experience	Diploma (Electrical)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	19/07/2016
Next Review Date	31/03/2022
NSQC Approval Date	09/10/2017
Version	1.0
Reference code on NQR	2017/POW/PSSC/01995
NQR Version	1.0



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PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



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- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



Qualification Pack

PSS/N2001: Use basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace
- PC5.** follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),
- PC6.** follow warning signs (danger, out of service, etc.) while working with electrical systems
- PC7.** use standard safe working practices when working at heights, confined areas and trenches
- PC8.** test any electrical equipment and system using insulated testing devices before touching them
- PC9.** ensure positive isolation of electrical equipment & system as per given standards
- PC10.** recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed
- PC11.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC12.** state methods of accident prevention in the work environment of the job role
- PC13.** state location of general health and safety equipment in the workplace
- PC14.** inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder
- PC15.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa
- PC16.** inspect grid station and its equipment routinely for any signs of oil and water leakage
- PC17.** store flammable materials and machine lubricating oil safely and correctly
- PC18.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC19.** ensure proper working condition of battery and battery charger
- PC20.** maintain electrolyte level of each cell with distilled water
- PC21.** maintain proper ventilation in battery room
- PC22.** apply good housekeeping practices at all times
- PC23.** identify common hazard signs displayed in various areas



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PC24. retrieve and/or point out documents that refer to health and safety in the workplace

PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly

Fire Safety

To be competent, the user/individual on the job must be able to:

PC26. use the various appropriate fire extinguishers on different types of fires correctly

PC27. distinguish between various types of fire

PC28. demonstrate rescue techniques applied during fire hazard

PC29. demonstrate good housekeeping in order to prevent fire hazards

PC30. demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC31. demonstrate how to free a person from electrocution

PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC33. demonstrate basic techniques of bandaging

PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC37. demonstrate the artificial respiration and the cpr process

PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work

PC39. write accident/incident report or dictate a report to another person, and send report to person responsible

PC40. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. escalation matrix and procedures for reporting work and employment related issues

KU5. meaning of hazards and risks

KU6. health and safety hazards commonly present in the work environment and related precautions

KU7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible



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- KU8.** possible causes of risk and accident
- KU9.** methods of accident prevention
- KU10.** safe working practices when working with tools and machines
- KU11.** safe working practices while working at various hazardous sites
- KU12.** where to find all the general health and safety equipment in the workplace
- KU13.** various dangers associated with the use of electrical equipment
- KU14.** positive isolation of electrical equipment and system
- KU15.** safe handling and disposal of hazardous power plant wastes
- KU16.** use of emission and pollution control devices and measures taken to control pollution
- KU17.** various safety procedures and equipment used to work at heights, trenches and confined places
- KU18.** safe working practices specific to working with electrical equipment & system e.g. lock out/tag out, ptw, etc.
- KU19.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU20.** importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
- KU21.** precautionary activities taken to prevent fire accident
- KU22.** various causes of fire
- KU23.** techniques of using the different fire extinguishers
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire
- KU26.** emergency rescue techniques applied during a fire hazard
- KU27.** various types of safety signs and what they mean
- KU28.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge.
- GS2.** note down observations (if any) related to the operation/maintenance.
- GS3.** read and interpret the process required for different types of manuals for maintenance.
- GS4.** read and interpret the flowchart of all parts of an assembly.
- GS5.** read manuals and documents to understand the product-details & how they can be used.
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor.
- GS7.** effectively communicate with the team members.
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor.
- GS9.** communicate clearly with the colleague on the issues faced during query/fault.
- GS10.** follow colleague/contractor rule-based decision making process.
- GS11.** take decisions with systematic course of actions and/or response.



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- GS12.** planning and organization of tasks to meet deadlines.
- GS13.** build customer relationships and use customer centric approach.
- GS14.** seek and comprehend operation related inputs for clarification
- GS15.** find ways of modifying difficult operating stages to make it operation friendly
- GS16.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results.
- GS17.** quick approach and solution towards faults repairing.
- GS18.** critically evaluate operation parameters in relation to system normality
- GS19.** develop a holistic and comprehensive profile of grid station on segregated discrete process.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	11	46	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	-	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	-	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	-	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace	1	1	-	-
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),	1	2	-	-
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems	1	1	-	-
PC7. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC8. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC9. ensure positive isolation of electrical equipment & system as per given standards	1	2	-	-
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed	1	2	-	-
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	1	-	-
PC12. state methods of accident prevention in the work environment of the job role	-	2	-	-
PC13. state location of general health and safety equipment in the workplace	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder	-	2	-	-
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa	-	2	-	-
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage	-	2	-	-
PC17. store flammable materials and machine lubricating oil safely and correctly	-	2	-	-
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	1	-	-
PC19. ensure proper working condition of battery and battery charger	1	1	-	-
PC20. maintain electrolyte level of each cell with distilled water	-	2	-	-
PC21. maintain proper ventilation in battery room	-	1	-	-
PC22. apply good housekeeping practices at all times	1	2	-	-
PC23. identify common hazard signs displayed in various areas	-	2	-	-
PC24. retrieve and/or point out documents that refer to health and safety in the workplace	-	2	-	-
PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly	-	3	-	-
Fire Safety	4	9	-	-
PC26. use the various appropriate fire extinguishers on different types of fires correctly	1	1	-	-
PC27. distinguish between various types of fire	1	2	-	-
PC28. demonstrate rescue techniques applied during fire hazard	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate good housekeeping in order to prevent fire hazards	-	2	-	-
PC30. demonstrate the correct use of a fire extinguisher	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	21	-	-
PC31. demonstrate how to free a person from electrocution	1	2	-	-
PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	-	3	-	-
PC33. demonstrate basic techniques of bandaging	1	2	-	-
PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC37. demonstrate the artificial respiration and the cpr process	1	2	-	-
PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC39. write accident/incident report or dictate a report to another person, and send report to person responsible	1	2	-	-
PC40. demonstrate correct method to move injured people and others during an emergency	1	2	-	-
NOS Total	24	76	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2001
NOS Name	Use basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022



Qualification Pack

PSS/N3007: Carry out Installation in power distribution systems

Description

Junior engineer will undertake installation, testing & commissioning of 33/11 substation, distribution transformer, O/H Line, U/G cabling, distribution transformer, release of new connection, consumer management, GIS, SCADA, Metering, etc. as per standards and design drawing. . Periodic /preventive and breakdown maintenance of all the aforesaid.

Scope

This unit/task covers the following:

Elements and Performance Criteria

Power distribution system Installation

To be competent, the user/individual on the job must be able to:

- PC1.** apply understanding of power distribution system
- PC2.** apply knowledge of type of distribution systems with respect to voltage level, network configuration (ring main/radial etc.)
- PC3.** apply understanding of cables/conductors their size and specifications
- PC4.** carry out erection and commissioning of substation
- PC5.** carry out the route survey for O/H line or U/G cable distribution supply
- PC6.** carry out installation of distribution transformer
- PC7.** supervise erection of line poles, substation, o/h line or u/g cable, switchgear etc.
- PC8.** plan and execute service line connection for customers
- PC9.** ensure that all the tools & equipment needed for erection or installation are available at site
- PC10.** undertake meter installation at customer premises
- PC11.** apply knowledge of SCADA and GIS mapping
- PC12.** ensure proper earthing of equipment for healthy operation
- PC13.** undertake installation of protection devices- surge protection device, over voltage protection etc.
- PC14.** read and understand network schematic, line diagrams and related technical drawings
- PC15.** coordinate and manage all the logistics, material planning and handling related issues
- PC16.** monitor power supply from substation during work in progress
- PC17.** test and inspect transformer, switchgear etc. on post commissioning
- PC18.** have operational familiarity with tools and tackles
- PC19.** be responsible for mobilizing resources
- PC20.** coordinate with seniors and also monitor with workers/helpers

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** relevant legislation, standards, policies, and procedures followed in the organization relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in a clean and safe environment
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** common electricity terminology and correct interpretation of the same terminology: e.g. current, voltage, resistance, kilowatt (kw), kilowatt hour (kwh)
- KU10.** distribution system plant and equipment
- KU11.** entire value chain of distribution system
- KU12.** technical parameter and its function of power distribution equipment
- KU13.** smart grid, AMR, AMI and automation system
- KU14.** metering, billing and collection system also customer care service
- KU15.** importance of reporting problems in a timely manner
- KU16.** ratings and specifications of cables, fuses, switches and wires
- KU17.** handling all machineries, equipment & vehicles
- KU18.** appropriate judgment and initiative pertaining to work methods and tools
- KU19.** technical manuals, blueprints, schematics, diagrams, plans, specifications estimate time, material and equipment needed to complete assignments
- KU20.** quality parameters, quality assessment based on physical parameters
- KU21.** metering system and its installation
- KU22.** service line connection process

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the supervisor or engineer
- GS2.** note down observations (if any) related to the process
- GS3.** use basic IT skills including report preparation and data analysis
- GS4.** read and interpret the process required for various types of operations
- GS5.** read and interpret and process flowchart for all operations
- GS6.** read manuals and operation documents to understand the equipment used into operation
- GS7.** discuss task lists, schedules and activities with the supervisor
- GS8.** effectively communicate with the team members
- GS9.** communicate clearly with the customer on the issues faced during query/fault
- GS10.** follow organization rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response



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- GS12.** planning and organization of tasks to meet deadlines
- GS13.** record keeping, documentation
- GS14.** build customer relationships and use customer centric approach
- GS15.** seek and comprehend operation related inputs for clarification
- GS16.** find ways of modifying difficult operating stages to make it operation friendly
- GS17.** apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS18.** critically evaluate operation parameters in relation to product features intended
- GS19.** develop a holistic and comprehensive profile of products based on segregated discrete process stages



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Power distribution system Installation</i>	30	70	-	-
PC1. apply understanding of power distribution system	2	4	-	-
PC2. apply knowledge of type of distribution systems with respect to voltage level, network configuration (ring main/redial etc.)	2	4	-	-
PC3. apply understanding of cables/conductors their size and specifications	2	4	-	-
PC4. carry out erection and commissioning of substation	2	4	-	-
PC5. carry out the route survey for O/H line or U/G cable distribution supply	2	4	-	-
PC6. carry out installation of distribution transformer	2	4	-	-
PC7. supervise erection of line poles, substation, o/h line or u/g cable, switchgear etc.	2	4	-	-
PC8. plan and execute service line connection for customers	2	3	-	-
PC9. ensure that all the tools & equipment needed for erection or installation are available at site	1	3	-	-
PC10. undertake meter installation at customer premises	1	4	-	-
PC11. apply knowledge of SCADA and GIS mapping	2	4	-	-
PC12. ensure proper earthing of equipment for healthy operation	1	4	-	-
PC13. undertake installation of protection devices- surge protection device, over voltage protection etc.	1	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. read and understand network schematic, line diagrams and related technical drawings	2	4	-	-
PC15. coordinate and manage all the logistics, material planning and handling related issues	1	3	-	-
PC16. monitor power supply from substation during work in progress	1	3	-	-
PC17. test and inspect transformer, switchgear etc. on post commissioning	1	3	-	-
PC18. have operational familiarity with tools and tackles	1	3	-	-
PC19. be responsible for mobilizing resources	1	3	-	-
PC20. coordinate with seniors and also monitor with workers/helpers	1	2	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3007
NOS Name	Carry out Installation in power distribution systems
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	19/07/2016
Next Review Date	31/03/2022
NSQC Clearance Date	09/10/2017



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PSS/N3008: Carry out operation and maintenance of power distribution system

Description

This section covers the operation and maintenance of distribution system like transformer, substation, O/H line, U/G cable, switchgear, service cable, meters and associated components etc. while ensuring healthy operation of distribution system equipment.

Scope

This unit/task covers the following:

Elements and Performance Criteria

Primary Inspection for maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** inspect substation equipment, power transformer, distribution transformer, switchgear, overhead lines, insulators and other related equipment for identification of faults, possible wear and tear and to assess requirement of proactive preventive maintenance and breakdown maintenance on need basis
- PC2.** carry out/monitor/supervise maintenance related activities pertaining to equipment installed in sub stations
- PC3.** apply understanding of revenue process management viz. release of new connection, meter installation, meter reading, bill generation, bill distribution, revenue collection
- PC4.** apply understanding of various consumer categories and applicable tariffs
- PC5.** carry out/monitor/supervise maintenance of O/H line and U/G cable
- PC6.** check all the intersections & joints (termination) in the wiring or cable
- PC7.** check the running parameter of distribution system as per design standard
- PC8.** monitor working condition of transformer (overloading/under loading) and other equipment
- PC9.** locate the conduit, cables & other undergoing devices to perform maintenance work
- PC10.** apply understanding of metering technologies (electronic meter, automated meter reading, smart meter etc.)
- PC11.** monitor performance of critical system such as remote terminal units, remote metering units and other automation system
- PC12.** carry out routine maintenance

Testing of the system

To be competent, the user/individual on the job must be able to:

- PC13.** carry out all the testing equipment like tester, multimeter, control cable etc.
- PC14.** test the system parameter to know abnormal condition of the system
- PC15.** test the healthiness of connected equipment
- PC16.** maintain log of system condition (parameters)
- PC17.** undertake software upgradation and testing



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Repairing & replacement

To be competent, the user/individual on the job must be able to:

- PC18.** carry out repair and replacement of faulty/ unhealthy equipment
- PC19.** troubleshoot faulty system
- PC20.** upgrade or modify the existing unhealthy equipment/system
- PC21.** carry out general routine repair work
- PC22.** implement technical change in equipment/system

Activity specific safety

To be competent, the user/individual on the job must be able to:

- PC23.** use of PPE: e.g. safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), safety rope ,ladder etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the organization relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in a clean and safe environment
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** common electricity terminology and correct interpretation of the same terminology: e.g. current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
- KU10.** various test/inspection of power distribution system equipment
- KU11.** routine and preventive maintenance of distribution system equipment
- KU12.** o&m process of each equipment of distribution system
- KU13.** automation system , AMR and AMI
- KU14.** smart grid, knowledge of SCADA and other communication system
- KU15.** importance of reporting problems in a timely manner
- KU16.** ratings and specifications of cables, fuses, switches and wires
- KU17.** handling of all machineries, equipment & vehicles
- KU18.** appropriate judgment and initiative pertaining to work methods and tools
- KU19.** technical manuals, blueprints, schematics, diagrams, plans, specifications estimate time, material and equipment needed to complete assignments
- KU20.** quality parameters, quality assessment based on physical parameters

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the supervisor or engineer
- GS2.** note down observations (if any) related to the process of installation, operation and maintenance of power distribution system
- GS3.** read and interpret the process required for various types of operations related to power distribution system
- GS4.** read and interpret and process flowchart for all operations related to power distribution system
- GS5.** read manuals and operation documents to understand the equipment used into operation
- GS6.** discuss task lists, schedules and activities with the supervisor
- GS7.** effectively communicate with the team members
- GS8.** communicate clearly with the customer on the issues faced during query/fault
- GS9.** follow organization rule-based decision making process
- GS10.** take decisions with systematic course of actions and/or response
- GS11.** planning and organization of tasks to meet deadlines
- GS12.** build customer relationships and use customer centric approach
- GS13.** seek and comprehend operation related inputs for clarification
- GS14.** find ways of modifying difficult operating stages to make it operation friendly
- GS15.** apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS16.** critically evaluate operation parameters in relation to product features intended
- GS17.** develop a holistic and comprehensive profile of products based on segregated discrete process

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Primary Inspection for maintenance</i>	19	37	-	-
PC1. inspect substation equipment, power transformer, distribution transformer, switchgear, overhead lines, insulators and other related equipment for identification of faults, possible wear and tear and to assess requirement of proactive preventive maintenance and breakdown maintenance on need basis	2	3	-	-
PC2. carry out/monitor/supervise maintenance related activities pertaining to equipment installed in sub stations	2	3	-	-
PC3. apply understanding of revenue process management viz. release of new connection, meter installation, meter reading, bill generation, bill distribution, revenue collection	2	3	-	-
PC4. apply understanding of various consumer categories and applicable tariffs	2	3	-	-
PC5. carry out/monitor/supervise maintenance of O/H line and U/G cable	2	3	-	-
PC6. check all the intersections & joints(termination) in the wiring or cable	2	3	-	-
PC7. check the running parameter of distribution system as per design standard	2	3	-	-
PC8. monitor working condition of transformer(overloading/under loading) and other equipment	1	4	-	-
PC9. locate the conduit, cables & other undergoing devices to perform maintenance work	1	3	-	-
PC10. apply understanding of metering technologies (electronic meter, automated meter reading, smart meter etc.)	1	3	-	-
PC11. monitor performance of critical system such as remote terminal units , remote metering units and other automation system	1	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. carry out routine maintenance	1	3	-	-
<i>Testing of the system</i>	5	15	-	-
PC13. carry out all the testing equipment like tester, multimeter, control cable etc.	1	3	-	-
PC14. test the system parameter to know abnormal condition of the system	1	3	-	-
PC15. test the healthiness of connected equipment	1	3	-	-
PC16. maintain log of system condition (parameters)	1	3	-	-
PC17. undertake software upgradation and testing	1	3	-	-
<i>Repairing & replacement</i>	5	15	-	-
PC18. carry out repair and replacement of faulty/unhealthy equipment	1	3	-	-
PC19. troubleshoot faulty system	1	3	-	-
PC20. upgrade or modify the existing unhealthy equipment/system	1	3	-	-
PC21. carry out general routine repair work	1	3	-	-
PC22. implement technical change in equipment/system	1	3	-	-
<i>Activity specific safety</i>	1	3	-	-
PC23. use of PPE: e.g. safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), safety rope ,ladder etc.	1	3	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3008
NOS Name	Carry out operation and maintenance of power distribution system
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	19/07/2016
Next Review Date	31/03/2022
NSQC Clearance Date	09/10/2017

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2001.Use basic health and safety practices for power related work	24	76	-	-	100	15
PSS/N3007.Carry out Installation in power distribution systems	30	70	-	-	100	40
PSS/N3008.Carry out operation and maintenance of power distribution system	30	70	-	-	100	40
Total	114	286	-	-	400	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PPE	Personal Protective Equipment
KW	Kilowatt
KV	Kilo Volt
KWH	Kilo Watt Hour
PTW	Permit to work
CPR	Cardio Pulmonary Resuscitation
GIS	Geographical Information System
RTU	Remote Terminal Unit
CBIP	Central Board of Irrigation and Power
CEA	Central Electricity Authority
CERC	Central Electricity Regulatory Commission
CPRI	Central Power Research Institute
CT	Current Transformer
DC	Direct Current
DISCOM	Distribution Company
DT	Distribution Transformer
E/F	Earth Fault
HT	High Tension
HVDS	High Voltage Distribution System
IE Act	Indian Electricity Act 2003
IS	Indian Standard



Qualification Pack

LA	Lightening Arrestor
MD	Maximum Demand
MVA	Mega Volt Ampere
MW	Mega Watt
MWh	Mega Watt hour
O/C	Over Current
O/H	Over Head
O&M	Operation & Maintenance
OPGW	Optical Ground Wire
PT	Potential Transformer
RMU	Ring Main Unit
SCADA	Supervisory Control and Data Acquisition
SEB	State Electricity Board
SERC	State Electricity Regulatory Commission
SMS	Short Message Service
T&D	Transmission and Distribution
T/F	Transformer
VT	Voltage Transformer
XLPE	Cross Linked Poly Ethylene Cable

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.